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Machine learning derivation of two cardiac arrest subphenotypes with distinct responses to treatment

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Abstract

Introduction Cardiac arrest (CA), characterized by its heterogeneity, poses challenges in patient management. This study aimed to identify clinical subphenotypes in CA patients to aid in patient classification, prognosis assessment, and treatment decision-making.

Methods For this study, comprehensive data were extracted from the Medical Information Mart for Intensive Care IV (MIMIC-IV) 2.0 database. We excluded patients under 18 years old, those not initially admitted to the intensive care unit (ICU), or treated in the ICU for less than 72 h. A total of 57 clinical parameters relevant to CA patients were selected for analysis. These included demographic data, vital signs, and laboratory parameters. After an extensive literature review and expert consultations, key factors such as temperature (T), sodium (Na), creatinine (CR), glucose (GLU), heart rate (HR), PaO₂/FiO₂ ratio (P/F), hemoglobin (HB), mean arterial pressure (MAP), platelets (PLT), and white blood cell count (WBC) were identified as the most significant for cluster analysis. Consensus cluster analysis was utilized to examine the mean values of these routine clinical parameters within the first 24 h post-ICU admission to categorize patient classes. Furthermore, in-hospital and 28-day mortality rates of patients across different CA subphenotypes were assessed using multivariate logistic and Cox regression analysis.

Results After applying exclusion criteria, 719 CA patients were included in the study, with a median age of 67.22 years (IQR: 55.50–79.34), of whom 63.28% were male. The analysis delineated two distinct subphenotypes: Subphenotype 1 (SP1) and Subphenotype 2 (SP2). Compared to SP1, patients in SP2 exhibited significantly higher levels of P/F, HB, MAP, PLT, and Na, but lower levels of T, HR, GLU, WBC, and CR. SP2 patients had a notably higher in-hospital mortality rate compared to SP1 (53.01% for SP2 vs. 39.36% for SP1, $P < 0.001$). 28-day mortality decreased continuously for both subphenotypes, with a more rapid decline in SP2. These differences remained significant after adjusting for potential covariates (adjusted OR = 1.82, 95% CI: 1.26–2.64, $P = 0.002$; HR = 1.84, 95% CI: 1.40–2.41, $P < 0.001$).

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Conclusions The study successfully identified two distinct clinical subphenotypes of CA by analyzing routine clinical data from the first 24 h following ICU admission. SP1 was characterized by a lower rate of in-hospital and 28-day mortality when compared to SP2. This differentiation could play a crucial role in tailoring patient care, assessing prognosis, and guiding more targeted treatment strategies for CA patients.

Keywords Cardiac arrest, Machine learning, Subphenotypes, Precision medicine, Critically illness, Latent class analysis, Mortality

Introduction

The latest statistics from the American Heart Association (AHA) indicate that the incidence of out-of-hospital cardiac arrest (OHCA) stands at 140.7 cases per 100,000 population, while in-hospital cardiac arrest (IHCA) occurs at a rate of 17.16 cases per 1,000 hospitalized patients [1]. Cardiac arrest (CA) is a significant health concern, causing nearly 370,000 deaths annually in the United States [1] and almost 1 million deaths each year in China [2]. These numbers have been rising, especially in the period following the COVID-19 pandemic [1]. The high mortality rate associated with CA is largely attributed to its heterogeneity, making the identification of specific CA subphenotypes critical for the development of precise and effective treatment plans.

Now, there remains ambiguity about the various terms created to classify and study CA, and consensus documents from various cardiovascular societies characterize them distinctively [3–5]. North American (American Heart Association/Heart Rhythm Society/American Council of Cardiology), European (European Heart Society), and Asian-Pacific (Asia-Pacific Heart Rhythm Society) all have unique definitions of CA [6]. They differed in whether the emphasis is placed on timing, etiology, or situational context surrounding the cardiac arrest. However, the above-mentioned traditional classification methods inevitably have the following problems: (1) Unable to accurately reflect the pathophysiological state: The pathophysiological state of each patient after CA can be significantly different, but these classification criteria do not accurately reflect these differences; (2) Inability to provide effective support for treatment decisions: Different types of cardiac arrest require different treatment strategies. The above classification methods ignore individual differences and cannot provide enough clinical information to guide treatment. Therefore, this study focused on the above two issues, hoping to use a new tool to derive a new clinical classification method for CA patients. This classification method does not rely on external factors such as timing, etiology, or situational context surrounding the CA, but uses demographic variables, severity scores, comorbidities, vital signs, laboratory parameters, medication, and special treatments to depict the high-latitude stereoscopic characteristics of each CA patient, and uses computer tools to classify them in high-latitude digital space. This

classification method is based on the pathophysiological state of CA patients, which can evaluate the prognosis of patients and help with clinical treatment decisions.

The machine learning (ML) classifier model has become one of the most indispensable tools in modern medical research for identifying various disease subphenotypes. Among them, consensus cluster analysis is a suitable clustering method that has been widely used to identify sepsis [7] and acute respiratory distress syndrome (ARDS) subphenotypes [8]. Until the initiation of this study, the application of similar methods for identifying clinical subphenotype models in CA patients had not been widely explored by researchers. This study sought to develop a consensus cluster analysis approach to identify clinical subphenotypes of CA patients based on data collected within 24 h of admission to the intensive care unit (ICU). This methodology aims to assist clinicians in classifying and assessing the prognosis of CA patients, thereby enabling the potential for early warning and precise interventions for CA in the future.

Methods

Data source

All the data were extracted from the Medical Information Mart for Intensive Care IV (MIMIC-IV) 2.0 database, which contained all medical record numbers corresponding to patients admitted to the ICU or emergency department between 2008 and 2019 in the Beth Israel Deaconess Medical Center (BIDMC). Access to the database was granted following the completion of NIH-required online training by one of the authors (Mengyuan Diao, with certification ID: 1630201). The Institutional Review Board (IRB) at BIDMC provided a waiver for informed consent and approved the use of this resource for research purposes. Consequently, the study was conducted using publicly available and anonymized data, negating the need for individual patient consent.

Study population and outcome

In this study, patients who experienced CA were included as research participants, regardless of the cause of the arrest. However, the study excluded patients who were under 18 years of age, as well as those who were either initially admitted to the ICU or treated in the ICU for less than 72 h. A flow diagram detailing the study design and participant selection is presented in Fig. 1. The primary

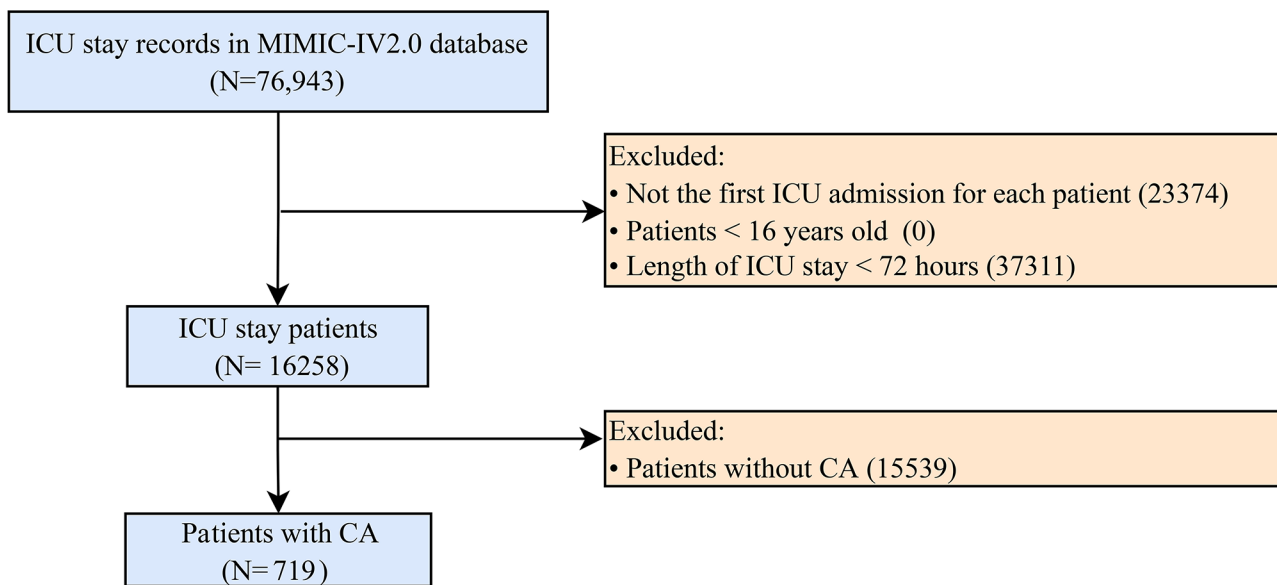


Fig. 1 Flow chart of patient selection. Ultimately, 719 patients with cardiac arrest (CA) from the intensive care unit (ICU) were enrolled in this study MIMIC-IV, Medical Information Mart for Intensive Care-IV; ICU, intensive care unit; CA, cardiac arrest

outcome measured in the study was in-hospital mortality, while the secondary outcome focused on the 28-day mortality.

Data extraction and variable selection

The extracted data for this study encompassed a comprehensive range of parameters including demographic variables, severity scores, comorbidities, vital signs, laboratory parameters, medication, and special treatments. The demographic variables considered were age, race, sex, and body mass index (BMI). Comorbidities included in the data were hypertension, heart failure, cerebral infarction, chronic obstructive pulmonary disease (COPD), cirrhosis of the liver, chronic kidney disease, malignant cancer, diabetes, myocardial infarction, arrhythmia, coronary heart disease, and cardiomyopathy. Vital signs incorporated into the study were heart rate (HR), systolic blood pressure (SBP), diastolic blood pressure (DBP), mean arterial pressure (MAP), respiratory rate (RR), temperature (T), $\text{PaO}_2/\text{FiO}_2$ (P/F), SPO_2 , and total input and output values. Laboratory parameters consisted of hemoglobin (HB), platelets (PLT), white blood cells (WBC), bicarbonate, chloride, creatinine (CR), glucose (GLU), sodium (Na), potassium, bilirubin, pH levels, PO_2 , PCO_2 , international normalized ratio (INR), base excess (BE), alanine aminotransferase (ALT), aspartate transaminase (AST), blood urea nitrogen (BUN), and lactate (LAC). Severity scoring included the Charlson score and the Sequential Organ Failure Assessment (SOFA) score. The medications considered were vasoactive drugs, antiarrhythmics, glucocorticoids, and sodium bicarbonate. Special treatments accounted

for were positive end-expiratory pressure (PEEP), percutaneous transluminal coronary intervention (PCI), extracorporeal membrane oxygenation (ECMO), continuous renal replacement therapy (CRRT), intra-aortic balloon pump (IABP), and mechanical ventilation. All these parameters were average values within the first 24 h post-ICU admission, except for input and output, which were cumulative totals within the same timeframe.

For feature selection, variables with a missing fraction exceeding 30% were excluded. Data were imputed with the use of multiple imputation for variables that were missing by less than 30%. The proportion of missing data is detailed in supplementary Fig. 1 (sFigure 1). Selection of variables was informed by prior literature and their potential association with the onset and progression of CA, as identified through literature review and expert discussions. The final variables chosen for consensus cluster analysis included T, Na, CR, GLU, HR, P/F, HB, MAP, PLT, and WBC. The consensus clustering analysis utilized the k-means clustering method as its internal algorithm, with each variable representing a different physiological system or function. Correlations between the selected parameters were generally low, as indicated in supplementary Fig. 2 (sFigure 2), with the highest correlation observed between HB and MAP ($r=0.4$).

Subphenotype classification

In this study, consensus cluster analysis was employed to identify clinical subphenotypes of CA in ICU patients. The analysis utilized a pre-defined subsampling parameter set at 80%, with 50 iterations. The potential number of clusters (k) was limited to a range from 2 to 6,

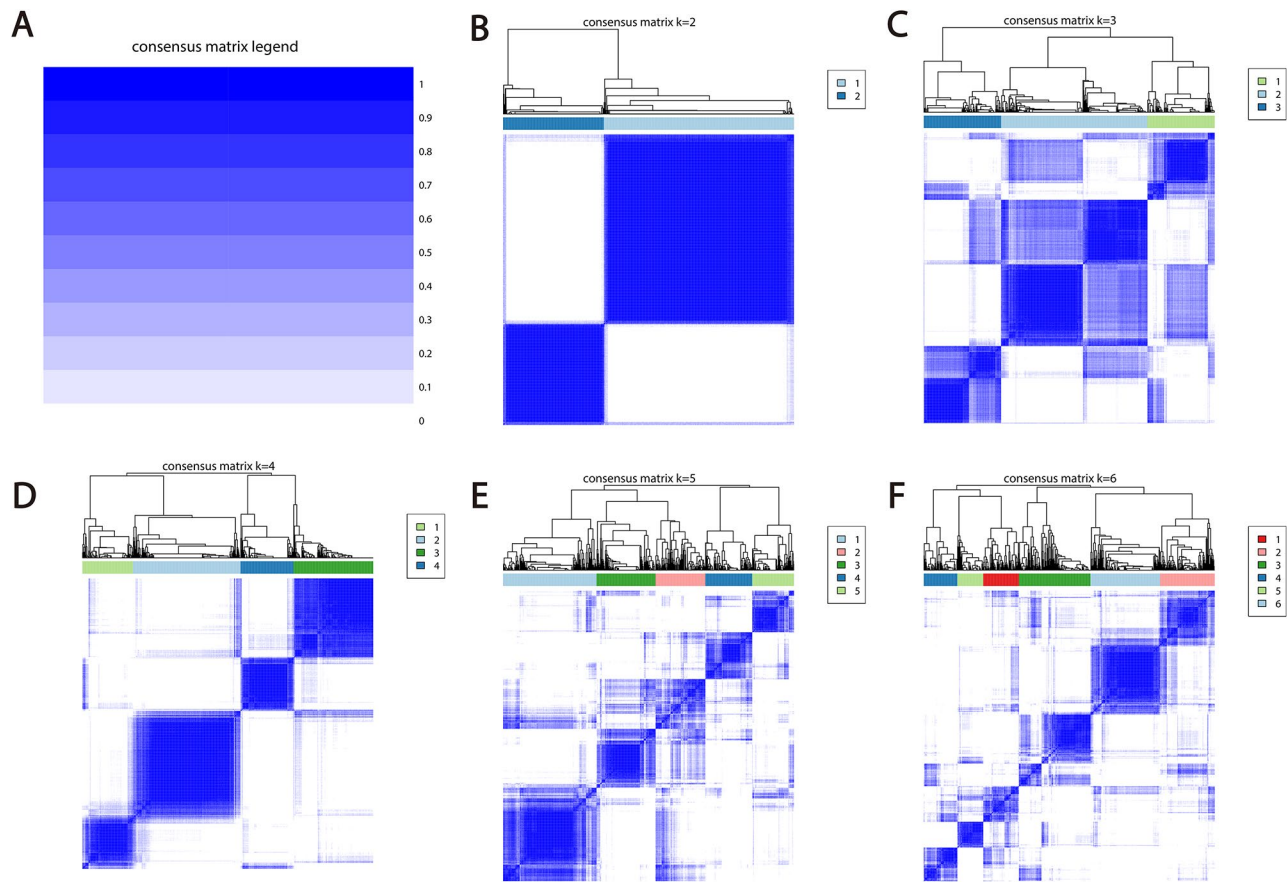


Fig. 2 (A) Consensus matrix legend. (B) Consensus matrix ($k=2$). (C) Consensus matrix ($k=3$). (D) Consensus matrix ($k=4$). (E) Consensus matrix ($k=5$). (F) Consensus matrix ($k=6$)

ensuring that the number of clusters remained clinically manageable and relevant. The optimal number of clusters was determined through a comprehensive examination of various metrics. These included the consensus matrix (CM) heat map, the cumulative distribution function (CDF), and cluster-consensus plots that focused on within-cluster consensus scores. The proportion of ambiguously clustered pairs (PAC) and the Bayesian information criterion (BIC) also played a crucial role in this determination. The within-cluster consensus score, which varies between 0 and 1, represented the average consensus value for all pairs of individuals within the same cluster. A score closer to 1 indicated greater stability and homogeneity within that cluster. PAC, calculated in the range of 0 to 0.9, measured the proportion of sample pairs with consensus values within pre-set boundaries. A lower PAC value signified improved cluster stability. Lastly, the BIC, ranging between 20,000 and 20,500, served as a criterion for the selection of the number of clusters, with lower BIC values indicating a more parsimonious model.

Statistical analysis

Statistical analyses in this study were conducted using the R software tool, version 4.1.2. After the CA patients clusters were identified, the differences between these clusters were analyzed. Measurement data were represented as medians with their interquartile ranges (IQR), and the nonparametric rank sum test was used to compare these measurements across different groups. Counting data, such as frequencies, were expressed in percentages, and the Chi-square test was employed to compare these frequencies between groups. For multivariable logistic regression, four distinct models were developed, each adjusted for different confounding variables. Model 1 served as the baseline with no adjustments. Model 2 included adjustments from Model 1, plus factors such as age and sex. Model 3 built upon Model 2 by also adjusting for the Charlson score. Finally, Model 4 included all adjustments from Model 3, with the addition of the SOFA score. The odds ratios (ORs) and their 95% confidence intervals (CIs) were calculated using logistic regression to assess the prognosis for different patient classes. In this analysis, a p -value of less than 0.05 (two-tailed) was considered to indicate statistical significance.

Results

Characteristics of the cohorts

In this study, a total of 76,540 ICU patients from the MIMIC-IV database were initially considered. After applying the exclusion criteria, 719 patients who experienced CA were ultimately enrolled. The patient cohort in this study had a median age of 67.2 years, with an interquartile range (IQR) of 55.5 to 79.3 years. Among these patients, 63.2% (455 out of 719) were male. The racial demographics were predominantly white (389/719, 54.1%), followed by black (81/719, 11.2%), and others (249/719, 34.6%). The most common comorbidities among the patients were hypertension (489/719, 68.0%), heart failure (265/719, 36.8%), and chronic kidney disease (176/719, 24.4%). These demographic details are further outlined in Table 1.

Regarding treatments within the first 24 h post-ICU admission, a significant proportion of the patients received vasoactive drugs (74.4%, 535/719), and nearly all were put on ventilation on the first day (99.0%, 712/719). Additionally, 29.7% (214/719) received antiarrhythmic drugs, and 29.2% (210/719) were administered sodium bicarbonate during the same period. The median Sequential Organ Failure Assessment (SOFA) score across the patient cohort was 7 (IQR: 4–9), indicative of a generally severe level of illness among these patients.

Among included patients, 74.4%(535/719) were administered vasoactive agents dose within 24 h of ICU admission, 99.0%(712/719) received mechanical ventilation, 29.7%(214/719) were given antiarrhythmic agents dose, and 29.2%(210/719) were treated with sodium bicarbonate. Across the dataset, Charlson score had a median score of 6 (IQR: 4–8), indicating a high risk of comorbidity.

Derivation of two subphenotypes

The CDF plot (sFigure 3A) shows the consensus distributions for each cluster, where the curve had negligible variation when $k=2$. The relative change in the area under the CDF curve is shown in the delta plot (sFigure 3B), revealing significant differences in area when $k=3$ or 4, indicating the relative increased area became considerably small. The mean cluster consensus score was comparable between a scenario of two clusters (sFigure 3C), and the two clusters showed favorably low PACs by the criteria (take $[0, 0.9]$ as the predetermined boundary, sFigure 3D). Then, we computed the optimal BIC values using Least Common Ancestors analysis (LCA), which was found to be minimum (20100) when $k=2$, indicating the model reached the optimal clustering number (sFigure 4). The consensus matrix (CM) heatmap (Fig. 2) shows that consensus cluster analysis identifies clusters 2 and 4 with clear boundaries, indicating good cluster stability over repeated iterations.

Through consensus clustering analysis, two distinct patient groups emerged from the parameters of CA patients within the first 24 h of their admission to the ICU. For simplicity, we referred to the two classes as subphenotype 1 (SP1) and subphenotype 2 (SP2), followed by creating 2D images using t-distributed stochastic neighbor embedding (t-SNE) to mark the differences between before and after clustering (Fig. 3A) for easier exploration and visualization of two subphenotypes. Abnormal clinical variables of the two subphenotypes are shown in Fig. 4.

Characteristics of two subphenotypes

Patients within both subphenotypes were of similar age. Among those with SP1, 57.6% were males. These patients also had lower Charlson Comorbidity Index and SOFA scores, and they required fewer vasoactive drugs and CRRT. In terms of laboratory parameters, patients in SP1 exhibited elevated levels of HB, Bicarbonate, Chloride, pH, PO_2 , and BE, contrasted with diminished levels of Potassium, Bilirubin, INR, and LAC. Similarly, 73.8% were male in SP2. However, patients in SP2 showed higher PEEP and a higher comorbidities burden. In addition, SP2 patients demonstrated elevated GLU, BUN, ALT and AST levels (Table 1). The distinct variables of both subphenotypes are depicted in Fig. 3B. In comparison to SP2, SP1 is distinguished by markedly higher levels of P/F, HB, MAP, PLT and Na, alongside lower values of T, HR, GLU, WBC and CR.

Clinical outcomes of two subphenotypes

As shown in Fig. 5A, SP1 had a lower proportion of in-hospital mortality than SP2, where KM survival curves showed that SP1 had a lower 28-day mortality than SP2 ($P<0.001$) (Fig. 5B). The univariate model 1 was adjusted three times by including different parameters and analyzed using multivariate logistic and COX regression analysis, respectively. Results showed that in either model, the in-hospital mortality (OR=1.82, 95% CI: 1.26–2.64, $p=0.002$) and 28-day mortality (HR=1.84, 95% CI: 1.40–2.41, $p<0.001$) of SP2 patients were higher than SP1 (Table 2).

We also analyzed the relationship between different vital signs and mortality (Table 3), and results showed that patients with $MAP \geq 65$ mmHg had lower mortality than those with $MAP < 65$ mmHg, independent of a whole group or in SP1 or SP2 individually. The in-hospital mortality and 28-day mortality of SP1 patients with temperature $32 \sim 36^\circ\text{C}$ were lower than those with temperature $36 \sim 37.5^\circ\text{C}$ (OR=2.91, 95% CI: 1.46–5.78], HR=2.37 [95% CI: 1.52–3.69]). However, the analyses for SP2 patients were not significantly different for patients with temperature $36 \sim 37.5^\circ\text{C}$ and others. In the whole cohort, patients also tended to have a lower mortality

Table 1 Baseline characteristics between different Subphenotypes (median [Q1, Q3]/ n [%])

Variables	Total (n = 719)	Subphenotype 1 (n = 470)	Subphenotype 2 (n = 249)	p	statistic
Demographic variables					
Age	67.221 (55.509, 79.34)	66.695 (54.815, 79.3)	67.594 (56.261, 79.337)	0.478	-0.71
Sex, male	455 (63.282)	271 (57.66)	184 (73.896)	< 0.001	17.774
Race, n (%)				0.207	3.152
black	81 (11.266)	46 (9.787)	35 (14.056)		
white	389 (54.103)	256 (54.468)	133 (53.414)		
other	249 (34.631)	168 (35.745)	81 (32.53)		
BMI	27.758 (24.504, 32.162)	27.618 (24.07, 31.867)	28.395 (25.381, 33.333)	0.014	-2.469
Comorbidities					
Hypertension, n (%)	489 (68.011)	305 (64.894)	184 (73.896)	0.017	5.656
Heart failure, n (%)	265 (36.857)	154 (32.766)	111 (44.578)	0.002	9.258
Cerebral infarction, n (%)	80 (11.127)	56 (11.915)	24 (9.639)	0.424	0.638
COPD, n (%)	114 (15.855)	79 (16.809)	35 (14.056)	0.393	0.729
Cirrhosis of liver, n (%)	27 (3.755)	10 (2.128)	17 (6.827)	0.003	8.689
Chronic kidney disease, n (%)	176 (24.478)	48 (10.213)	128 (51.406)	< 0.001	147.182
Malignant cancer, n (%)	75 (10.431)	43 (9.149)	32 (12.851)	0.156	2.008
Diabetes, n (%)	236 (32.823)	130 (27.66)	106 (42.57)	< 0.001	15.743
Myocardial infarction, n (%)	213 (29.624)	122 (25.957)	91 (36.546)	0.004	8.253
Arrhythmia, n (%)	600 (83.449)	392 (83.404)	208 (83.534)	1	< 0.001
Coronary heart disease, n (%)	347 (48.261)	213 (45.319)	134 (53.815)	0.037	4.371
Cardiomyopathy, n (%)	72 (10.014)	47 (10)	25 (10.04)	1	< 0.001
Severity scores					
Charlson score	6 (4, 8)	5 (4, 7)	7 (5, 9)	< 0.001	-7.15
SOFA	7 (4, 9)	5 (3, 8)	9 (7, 11)	< 0.001	-12.41
Vital signs					
Temperature	36.746 (36.234, 37.154)	36.763 (36.236, 37.175)	36.714 (36.24, 37.097)	0.838	0.204
Heart rate	81.88 (70.212, 95.651)	80.58 (68.613, 92.982)	84.22 (72.958, 99.438)	0.002	-3.141
Respiratory rate	20.122 (17.565, 23.575)	19.561 (17.101, 22.645)	21.379 (18.293, 24.875)	< 0.001	-4.143
MAP	78.219 (71.845, 84.728)	79.321 (72.876, 86.13)	74.852 (69.609, 82.232)	< 0.001	4.938
SBP	112.4 (105.069, 122.858)	113.19 (106.026, 124.572)	110.417 (103.409, 119.333)	0.001	3.198
DBP	62.692 (55.506, 69.929)	63.698 (57.122, 71.012)	60.097 (53.821, 68.5)	0.001	3.417
PaO ₂ /FiO ₂	229.3 (145.7, 333.6)	249.9 (164, 356.4)	181.4 (120.2, 265.3)	< 0.001	6.39
spo ₂	98 (96.37, 99.197)	98.217 (96.767, 99.396)	97.484 (95.783, 98.889)	< 0.001	4.36
Input	6000 (3395, 9620)	5940 (3400, 9199.25)	6250 (3280, 10720)	0.269	-1.106
Output	1847 (1015.5, 3094)	2178.5 (1391, 3416.5)	1125 (502, 2125)	< 0.001	9.62
Laboratory parameters					
Hemoglobin	11 (9.4, 13.115)	11.5 (9.79, 13.388)	10.3 (8.65, 12.35)	< 0.001	5.384
WBC	13.05 (9.31, 17.45)	12.49 (9.232, 16.488)	14.8 (9.7, 19.1)	0.001	-3.303
Platelets	183.5 (138.07, 245.29)	194 (144, 248.625)	170 (125.25, 236.67)	0.001	3.209
INR	1.3 (1.15, 1.6)	1.25 (1.105, 1.465)	1.47 (1.2, 1.85)	< 0.001	-7.203
ALT	73 (32, 188.75)	63 (31, 163)	96.5 (38, 309)	< 0.001	-3.947
AST	129.5 (48, 307)	106 (45.5, 243)	182 (56.2, 448)	< 0.001	-4.345
Bilirubin	0.6 (0.4, 1.14)	0.6 (0.377, 0.95)	0.7 (0.4, 1.55)	< 0.001	-3.556
BUN	23.75 (16.708, 36.5)	19 (14.5, 24.65)	42 (29.5, 60)	< 0.001	-17.587
Creatinine	1.2 (0.83, 1.99)	0.95 (0.732, 1.2)	2.46 (1.93, 3.6)	< 0.001	-21.701
Glucose	153.6 (123.5, 201.25)	149.585 (122, 190.188)	161.2 (128.5, 224.4)	0.001	-3.282
Sodium	139 (136.5, 141.67)	139.33 (137.05, 141.95)	138.5 (135.33, 141)	0.001	3.317
Potassium	4.13 (3.85, 4.55)	4.05 (3.8, 4.358)	4.45 (4, 4.95)	< 0.001	-7.971
Chloride	105.33 (101.55, 108.5)	106 (103, 109.75)	103 (98.5, 107)	< 0.001	7.085
pH	7.35 (7.3, 7.4)	7.36 (7.31, 7.4)	7.32 (7.27, 7.38)	< 0.001	5.087
po ₂	122 (86.8, 164.105)	132 (94.082, 173.31)	106.29 (76.21, 143)	< 0.001	5.206
pco ₂	39 (34.805, 43.84)	39 (34.842, 43.53)	39.33 (34.78, 44.36)	0.875	-0.157
Bicarbonate	21.25 (18.5, 24)	22 (19.25, 24.5)	20 (16.75, 23)	< 0.001	5.425

Table 1 (continued)

Variables	Total (n = 719)	Subphenotype 1 (n = 470)	Subphenotype 2 (n = 249)	p	statistic
Base excess	-3 (-5.895, -0.15)	-2.29 (-4.8, 0)	-4.57 (-8, -0.75)	< 0.001	5.418
Lactate	2.4 (1.655, 3.8)	2.2 (1.6, 3.475)	2.929 (1.8, 5.142)	< 0.001	-4.975
Medication					
Vasoactive, n (%)	535 (74.409)	327 (69.574)	208 (83.534)	< 0.001	15.932
Antiarrhythmic, n (%)	214 (29.764)	143 (30.426)	71 (28.514)	0.654	0.2
Glucocorticoids, n (%)	26 (3.616)	21 (4.468)	5 (2.008)	0.141	2.164
Sodium bicarbonate, n (%)	210 (29.207)	100 (21.277)	110 (44.177)	< 0.001	40.182
Special treatments					
PEEP	5.96 (5, 9.38)	5.535 (5, 8.552)	7.57 (5.18, 10.32)	< 0.001	-4.911
PCI, n (%)	38 (5.285)	26 (5.532)	12 (4.819)	0.817	0.053
ECOM, n (%)	16 (2.225)	12 (2.553)	4 (1.606)	0.58	0.306
CRRT, n (%)	93 (12.935)	25 (5.319)	68 (27.309)	< 0.001	67.952
IABP, n (%)	38 (5.285)	20 (4.255)	18 (7.229)	0.128	2.312
Ventilation, n (%)	712 (99.026)	466 (99.149)	246 (98.795)	0.698	Fisher

Continuous variables were reported as medians (interquartile range [IQR]), categorical variables as count (percentage [%]). Abbreviations: BMI, body mass index; WBC, white blood cell; INR, international normalized ratio; ALT, alanine aminotransferase; AST, aspartate transaminase; BUN, blood urea nitrogen; SBP, systolic blood pressure; DBP, diastolic blood pressure; MAP, mean arterial pressure; PEEP, positive end-expiratory pressure; SOFA, sequential organ failure assessment; PCI, percutaneous transluminal coronary intervention; ECMO, extracorporeal membrane oxygenation; CRRT, continuous renal replacement therapy; IABP, intra-aortic balloon pump; COPD, chronic obstructive pulmonary disease

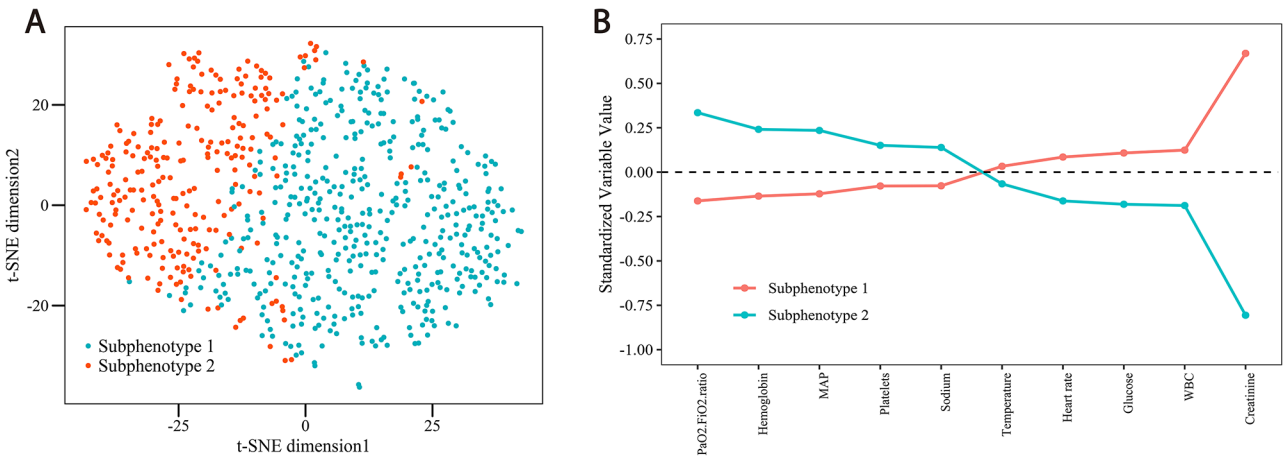


Fig. 3 (A) T-distributed stochastic neighbor embedding (t-SNE) plot. This nonlinear dimension reduction technique is used to visualize high-latitude data. (B) Selected variables by subphenotype in CA and the differences in the standardized values of each variable by subphenotype. All continuous variables were transformed into z-scores (mean: 0, standard deviation: -1 to 1) WBC, white blood cell count; MAP, mean arterial pressure

rate when $PO_2 \geq 80$ mmHg. Nevertheless, $PO_2 \geq 80$ mmHg showed the opposite outcome in SP2 as in SP1. Similarly, compared to patients with $PCO_2 < 35$ mmHg, those with $35 \text{ mmHg} \leq PCO_2 \leq 45 \text{ mmHg}$ or $PCO_2 \geq 45 \text{ mmHg}$ had lower mortality, independent of a whole group or in SP1 or SP2.

Discussion

In medicine, subphenotype and disease stage are two distinct concepts that can sometimes be difficult to distinguish. Is it possible that the discrimination that is being observed simply reflects two different stages of progression rather than separate clusters? Let me begin with our answer - “NO”. The two subphenotypes that have

been identified are clearly not two stages of CA progression, but rather refer to a group of CA patients who are further subdivided according to different characteristics or responses. The stages of CA progression can include prodromal phase, onset phase, CA phase, and biological death phase. The subphenotypes identified in this study are certainly in one of these stages of development, but it is more of a coexisting condition. The identification of subphenotypes usually relies on in-depth analysis of patient clinical characteristics (such as symptoms and signs) and biomarkers (such as specific molecules in blood and urine). These features and markers can help physicians classify patients into subgroups with similar

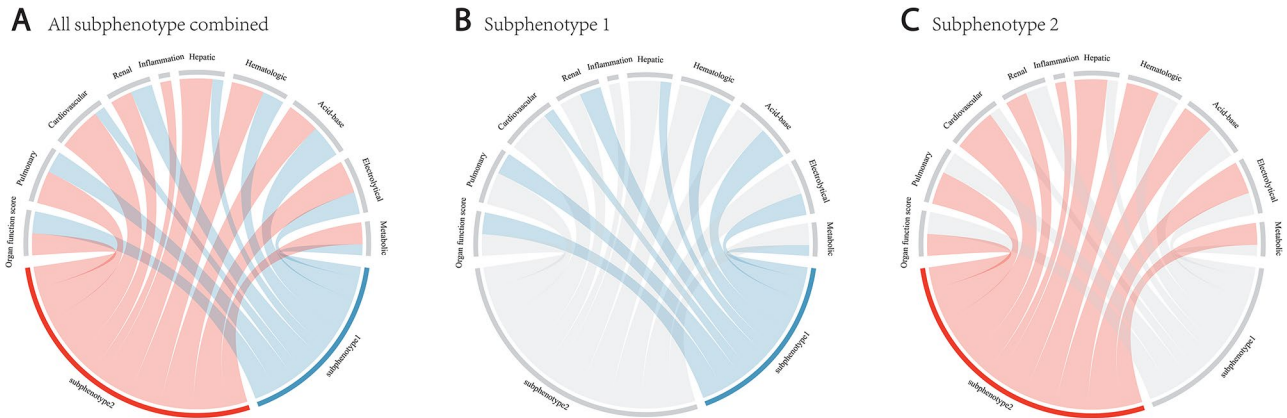


Fig. 4 Chord diagram showing abnormal clinical variables by suphenoype. In (A), the ribbons connect an individual subphenotype to an organ or system if the group mean is greater or less than the overall mean for the entire cohort. For example, subphenotype 1 (light blue) is more likely to have patients with acid–base imbalance (the ribbons connect to these portions of the circle) than patients with subphenotype 2 (light red), who are more likely to have cardiovascular, pulmonary, and hepatic dysfunction. In (B) and (C), each subphenotype is highlighted separately, and the ribbons connect to different patterns of clinical variables and organ or system dysfunctions located at the top of the circle

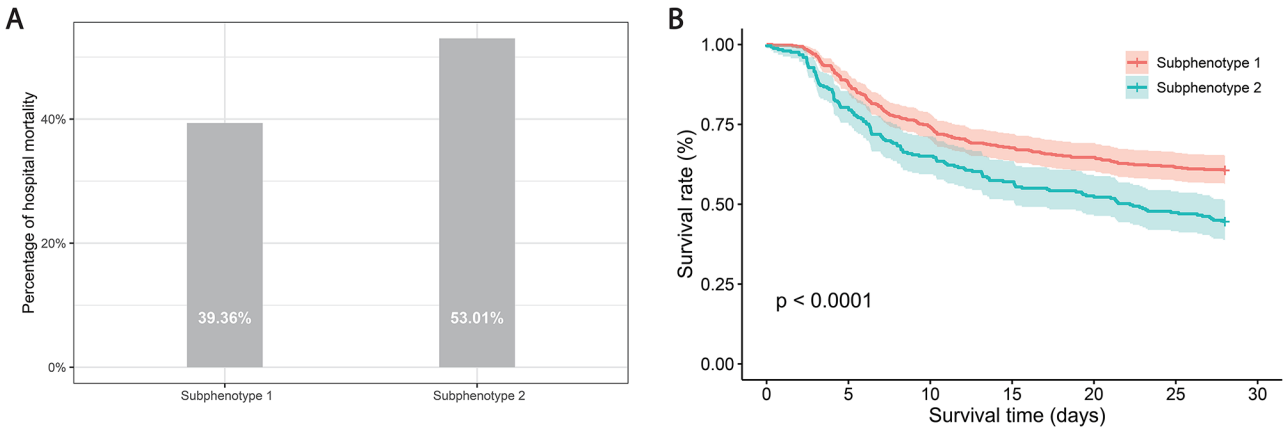


Fig. 5 (A) Hospital mortality among different subphenotypes of ICU patients with CA. (B) 28-day mortality among different subphenotypes of ICU patients with CA

Table 2 The associations of subphenotype and outcomes

	Hospital mortality			28-day mortality		
	OR	95%CI	P	HR	95%CI	P
Model 1						
Subphenotype 1	Ref			Ref		
Subphenotype 2	1.73	1.27–2.37	< 0.001	1.59	1.28–1.99	< 0.001
Model 2						
Subphenotype 1	Ref			Ref		
Subphenotype 2	1.83	1.33–2.52	< 0.001	1.64	1.31–2.05	< 0.001
Model 3						
Subphenotype 1	Ref			Ref		
Subphenotype 2	1.79	1.28–2.50	< 0.001	1.74	1.37–2.20	< 0.001
Model 4						
Subphenotype 1	Ref			Ref		
Subphenotype 2	1.82	1.26–2.64	0.002	1.84	1.40–2.41	< 0.001

Table 3 The associations of various vital signs and outcome in different populations

	Hospital mortality			28-day mortality		
	OR	95%CI	P	HR	95%CI	P
The whole cohort						
MAP						
< 65 mmHg	Ref			Ref		
≥ 65 mmHg	0.56	0.3–1.04	0.066	0.76	0.5–1.14	0.185
Temperature						
32–36 °C	Ref			Ref		
36–37.5 °C	2.73	1.56–4.77	< 0.001	2.13	1.52–3	< 0.001
other	1.35	0.88–2.06	0.166	1.33	0.98–1.81	0.068
PO ₂						
< 80 mmHg	Ref			Ref		
≥ 80 mmHg	0.86	0.56–1.33	0.509	0.85	0.62–1.15	0.293
PCO ₂						
< 35 mmHg	Ref			Ref		
35–45mmHg	0.46	0.3–0.69	< 0.001	0.61	0.46–0.8	< 0.001
≥ 45 mmHg	0.47	0.29–0.76	0.002	0.59	0.42–0.83	0.002
Subphenotype 1						
MAP						
< 65 mmHg	Ref			Ref		
≥ 65 mmHg	0.62	0.25–1.53	0.302	0.78	0.41–1.51	0.467
Temperature						
32–36 °C	Ref			Ref		
36–37.5 °C	2.91	1.46–5.78	0.002	2.37	1.52–3.69	< 0.001
other	1.54	0.92–2.57	0.100	1.47	0.99–2.16	0.053
PO ₂						
< 80 mmHg	Ref			Ref		
≥ 80 mmHg	0.89	0.48–1.63	0.699	0.94	0.59–1.5	0.802
PCO ₂						
< 35 mmHg	Ref			Ref		
35–45 mmHg	0.37	0.22–0.63	< 0.001	0.54	0.38–0.78	0.001
≥ 45 mmHg	0.43	0.23–0.8	0.007	0.54	0.35–0.83	0.005
Subphenotype 2						
MAP						
< 65 mmHg	Ref			Ref		
≥ 65 mmHg	0.58	0.23–1.43	0.233	0.82	0.48–1.4	0.461
Temperature						
32–36 °C	Ref			Ref		
36–37.5 °C	2.65	0.98–7.18	0.055	1.89	1.1–3.23	0.021
other	1.05	0.48–2.26	0.909	1.19	0.71–1.99	0.507
PO ₂						
< 80 mmHg	Ref			Ref		
≥ 80 mmHg	1.16	0.6–2.22	0.662	1.00	0.66–1.54	0.984
PCO ₂						
< 35 mmHg	Ref			Ref		
35–45 mmHg	0.71	0.36–1.37	0.306	0.74	0.49–1.12	0.153
≥ 45 mmHg	0.53	0.24–1.18	0.122	0.68	0.4–1.16	0.157

pathophysiological mechanisms or different responses to therapy.

The analysis of CA patient data identified two distinct clinical subphenotypes, referred to as SP1 and SP2. This was achieved through the use of consensus cluster analysis, employing a range of variables for the study, including

T, Na, CR, GLU, HR, P/F, HB, MAP, PLT, and WBC. Results showed that both subphenotypes differed in several dimensions, such as demographics, vital signs, laboratory results, and organ dysfunction, which differs from how patients are grouped using traditional methods. Furthermore, we refined the parameters of the univariate

model on three separate occasions. It was consistently observed that patients categorized under the SP2 had higher in-hospital and 28-day mortality rates compared to those with the SP1. The findings of this study are anticipated to serve as a vital reference for clinicians in classifying patients, assessing their prognosis, and making informed treatment decisions for CA patients.

ML has been widely used for classifying various disease subphenotypes, among which consensus cluster analysis is one of the most popular ML clustering algorithms, and has been widely used for identifying subphenotypes of diseases such as ARDS [8], sepsis [9], septic acute kidney injury (septic-AKI) [10], and liver dysfunction [11]. Under comprehensive consideration, consensus cluster analysis was selected for defining CA subphenotypes. CA subphenotypes can be determined after medical information collection and routine examination of patients following their admission, which is helpful for the timing of advanced treatment intervention, flexible adjustment of treatment plans, and screening of clinical trial subjects. After two critical care medicine experts discussed and reached a consensus, 52 relevant parameters were screened following the patient's admission to the ICU. To avoid excessive confounding leading to decreased clinical utility, we selected the most representative 10 parameters (T, Na, CR, GLU, HR, P/F, HB, MAP, PLT, and WBC) for cluster analysis. Only the mean values within 24 h after ICU admission were selected to achieve the early differentiation of clinical subphenotypes of CA patients. Although previous studies have shown that some new parameters can achieve considerable prediction results, it wasn't easy to carry out in practical clinical work [8, 12]. To achieve the study objective, clinically available data was used in the cluster model and two subphenotypes were derived. In these two phenotypes, SP2 patients were most closely associated with abnormalities in organs or systems, notably including the cardiovascular and hematologic systems, as well as hepatic dysfunction. Conversely, SP1 patients exhibited considerably more favorable characteristics than SP2. Since the statistical analysis showed that SP1 patients had overall fewer clinical anomalies despite a high proportion of acid-base imbalance, we may be able to classify patients with CA more quickly based on limited laboratory tests. The multiple cluster analyses showed consistency across clinical subphenotypes, which could help identify patients who would benefit from future intervention. Nonetheless, further understanding of the pathophysiological mechanisms underlying CA subphenotypes is necessary.

Among the 10 most representative parameters, there were 5 parameters which were higher or lower than SP2 patients, respectively. Results showed that patients with SP2 had higher HR and lower MAP than those with SP1, in addition to significant history of underlying

hypertension, heart failure, and a higher proportion of vasoactive drug use. A higher MAP within a specific range refers to better perfusion, leading to a better prognosis. A study recently reported an insignificant difference in outcome between the MAP of patients with CA between 77mmHg and 63mmHg [13]. However, the grouping method used in above study was fundamentally different from the subphenotypes classification method used in our study. Patients were grouped based on different target blood pressure in above study and ML was applied to cluster and classify patients under multi-parameter conditions in our study, which could be the primary reason why the results of the two studies are partially biased. Additionally, patients with potentially poor long-term outcomes were excluded from the sample. Hence, the influence of MAP on CA patients' outcome remains to be determined. Na was used in this study as a representative of electrolytes. The Na levels in both subphenotypes were similar and within the normal range but were slightly lower in SP2 patients than in SP1. Moreover, hypo- and hypernatremia were associated with a decreased probability of favorable neurological outcomes compared with normal Na [14]. It should also be noted that SP2 patients had lower Na even though they used sodium bicarbonate at a higher rate than SP1, suggesting that Na in SP2 patients might be lower before the use of sodium bicarbonate, and the difference between the two groups may be more significant. The main function of PLT is coagulation and hemostasis, and they are often used to evaluate the coagulation function of patients in clinic. The PLT of two subphenotypes in this study was within the normal range but slightly reduced in SP2 than in SP1. According to the neurologic outcome at 6 months post-CA, patients with good prognosis had a mean PLT of $230.31 \times 10^9/L$ at admission, and patients with poor prognosis had a mean PLT of $197.30 \times 10^9/L$ at admission [15], which were similar to our results and end outcomes. P/F is an important indicator of respiratory function and is widely used to distinguish high-risk patients with adverse clinical outcomes [16]. Our results showed that SP2 patients had lower P/F than SP1, suggesting that SP2 patients had poor respiratory function and more severe organ or tissue ischemia and hypoxia. Multiple factors could contribute to a lower P/F in CA patients, such as aspiration pneumonia, pulmonary embolism, systemic inflammatory reaction, pulmonary exudation, and ARDS, among which some were also common CA causes.

Furthermore, our results showed that patients with SP2 had abnormally high levels of CR, while patients with SP1 had normal levels of CR, which suggesting worse renal function in SP2 patients. Meanwhile, SP2 patients had a higher proportion of chronic kidney disease comorbidities. A decrease in CR of >0.2 mg/dl in the first 24 h may indicate a good prognosis, while a constant or even

elevated serum CR level suggests a poor prognosis [17]. Excessively high GLU can lead to immunosuppression and oxidative stress, leading to poor patient outcomes. A study reported that higher GLU levels were associated with poor neurological outcomes in those patients with CA treated with targeted temperature management (TTM) [18]. Our data showed that SP2 patients had higher GLU than SP1. The statistical analysis revealed infection as the most significant difference between the two subphenotypes since SP2 patients had a higher WBC than SP1, indicating a severe infection. However, it could have happened before CA onset, or the infection could be the cause of CA. Furthermore, a study reported that WBC had no significant ability to distinguish infection in CA patients receiving TTM [19]. Although WBC showed apparent difference between the two subphenotypes, but its reliability needs to be demonstrated by more clinical practice in the future. The temperature between the two subphenotypes not statistically significant, which was related to the TTM commonly adopted in clinical practice. Despite some controversy, TTM at 32–36 °C for at least 24 h post-CA remained the primary neuroprotective approach following OHCA or IHCA, consistent with AHA recommendations [20–22]. The results of COX and multivariate logistic analysis in this study showed that, in the whole population, the in-hospital and 28-day mortality of patients with T of 32–36 °C were lower than those of 36–37.5 °C, which indicated the benefit of TTM for the prognosis of CA patients.

CA patients are classified into two subphenotypes (SP1 and SP2) on the basis of different clinical features and physiological responses (on the basis of 10 vital signs and laboratory parameters that represent the pathophysiology of CA). Compared with the previous classification methods, these subphenotypes are more helpful for doctors to diagnose, treat and predict the prognosis of patients more accurately. The variables used for clustering can be obtained quickly after the patient is admitted to the hospital, so that clinicians can quickly identify the patient's subphenotype (SP1 or SP2). Clinicians can then tailor treatment to patients on the basis of the pathophysiological features of the subphenotypes that we have so far clustered. SP1 is more likely to have patients with acid–base imbalance than SP2 patients, who are more likely to have cardiovascular, pulmonary, and hepatic dysfunction. Therefore, attention to acid-base balance (e.g., an aggressive adjunctive CRRT strategy) is warranted beyond usual care in SP1 patients. For SP2 patients, we should be alert to the occurrence of multiple organ dysfunction syndrome (MODS). This may be because: 1. Poor basic organ function; 2. CA has a greater impact on the organs of patients. Active intra-aortic balloon pump (IABP), extracorporeal membrane oxygenation (ECMO) and artificial liver support system (ALSS) may be helpful

for patients with organ support. In-hospital and 28-day mortality were lower with SP1 than with SP2. We can make a preliminary prognosis assessment of CA patients according to the classification of patients after admission. This can be used as a theoretical basis for prognosis prediction in conversation with the patient's family.

Translating theoretical research results into clinical practice is a significant challenge faced by the majority of researchers. While previous studies found some critical factors in subphenotypes identification for ARDS and septic AKI, but these indicators have not been popularized because of the difficulty in extracting these indicators in clinical practice [8, 12, 23]. However, this challenge also exists in the classification of CA subphenotypes. Unlike previous research [12, 23], we used 10 conventional clinical variables to derive clinical CA subphenotypes, which were more straightforward and easier to obtain, and their values reflected the functional states of different systems or organs, making our cluster analysis more representative and universal. In the future, we aimed to conduct an external multicenter validation to refine the underlying model.

This study had several limitations. Firstly, since CA causes are numerous and complex, using the currently known disease patterns to identify all subphenotypes might not be sufficient. We should examine the heterogeneity of CA from a higher latitude. This puts forward higher requirements for researchers' ability to multi-system joint thinking. Secondly, the two experts believed that age is an important factor affecting the survival rate of patients with CA since younger tends to mean fewer underlying diseases, stronger immunity, and a better prognosis. Considering that the potential influence of age on patients with CA is fundamental and multifaceted, this study was not temporarily included. Thirdly, including more potential variables, such as shock type and microbiological data et cetera could provide new insights. However, these were not present in the MIMIC-IV 2.0 database. Lastly, the variables for clinical subphenotypes in this study were derived from a single-center retrospective database in the United States. Therefore, whether these subphenotypes can be generalized to more diverse populations of severely ill patients with CA in other parts of the world remains to be seen. We would invest more time and energy to verify and adjust it further. In the next few years, we planned to collect patients with CA information from multiple centers to establish the CA database of ICU patients in China.

Conclusions

Analysis of CA patients data retrieved from the MIMIC-IV 2.0 database revealed two clinical subphenotypes of CA, namely SP1 and SP2. Consensus cluster analysis was performed using the mean values of clinical, vital signs,

and laboratory indicators as the analysis variables. The SP1 ad considerably higher levels of P/F, HB, MAP, PLT, and Na, and lower T, HR, GLU, WBC, and CR levels than SP2. The in-hospital and 28-day mortality of patients with SP2 was higher than patients with SP1. The study outcomes are envisaged to provide helpful information to the clinicians regarding patient classification, prognosis evaluation, and making treatment decisions for prospective CA patients.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12967-024-05975-1>.

Supplementary Material 1

Supplementary Material 2

Supplementary Material 3

Supplementary Material 4

Supplementary Material 5

Supplementary Material 6

Author contributions

The specific division of labor was as follows: Conception, WZ; Funding, WH, and MD; Investigations, WZ; Methods, WZ, CW, and PN; Project management, SZ, HZ, and YZ.

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Declarations

Conflict of interest

The authors declare that there are no competing interests.

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References

1. Tsao CW, et al. Heart disease and stroke statistics-2022 update: a report from the American Heart Association. *Circulation*. 2022;145(8):e153–639.

2. Gu XM, et al. Meta-analysis of the success rate of heartbeat recovery in patients with prehospital cardiac arrest in the past 40 years in China. *Mil Med Res*. 2020;7(1):34.
3. Al-Khatib SM, et al. 2017 AHA/ACC/HRS guideline for management of patients with ventricular arrhythmias and the prevention of sudden cardiac death: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines and the Heart Rhythm Society. *Circulation*. 2018;138(13):e272–391.
4. Priori SG et al. 2015 ESC guidelines for the management of patients with ventricular arrhythmias and the prevention of sudden cardiac death: the task force for the management of patients with ventricular arrhythmias and the prevention of sudden cardiac death of the European Society of Cardiology (ESC). Endorsed by: Association for European Paediatric and Congenital Cardiology (AEPC). *Eur Heart J*. 2015;36(41):2793–2867.
5. Stiles MK, et al. 2020 APHRS/HRS expert consensus statement on the investigation of decedents with sudden unexplained death and patients with sudden cardiac arrest, and of their families. *Heart Rhythm*. 2021;18(1):e1–50.
6. Elfassy MD, et al. Understanding etiologies of cardiac arrest: seeking definitional clarity. *Can J Cardiol*. 2022;38(11):1715–8.
7. Seymour CW, et al. Derivation, validation, and potential treatment implications of novel clinical phenotypes for sepsis. *JAMA*. 2019;321(20):2003–17.
8. Maddali MV, et al. Validation and utility of ARDS subphenotypes identified by machine-learning models using clinical data: an observational, multicohort, retrospective analysis. *Lancet Respir Med*. 2022;10(4):367–77.
9. Hu C, et al. Application of machine learning for clinical subphenotype identification in sepsis. *Infect Dis Ther*. 2022;11(5):1949–64.
10. Chaudhary K, et al. Utilization of deep learning for subphenotype identification in sepsis-associated acute kidney injury. *Clin J Am Soc Nephrol*. 2020;15(11):1557–65.
11. Miao H, et al. Identification of subphenotypes of sepsis-associated liver dysfunction using cluster analysis. *Shock*. 2023;59(3):368–74.
12. Wiersema R, et al. Two subphenotypes of septic acute kidney injury are associated with different 90-day mortality and renal recovery. *Crit Care*. 2020;24(1):150.
13. Roedl K, Kluge S. Blood-pressure targets in comatose survivors of cardiac arrest. *N Engl J Med*. 2023;388(3):285.
14. Shida H, et al. Early prognostic impact of serum sodium level among out-of-hospital cardiac arrest patients: a nationwide multicentre observational study in Japan (the JAAM-OHCA registry). *Heart Vessels*. 2022;37(7):1255–64.
15. Kim HJ, et al. Time course of platelet counts in relation to the neurologic outcome in patients undergoing targeted temperature management after cardiac arrest. *Resuscitation*. 2019;140:113–9.
16. Villar J, et al. A universal definition of ARDS: the PaO₂/FiO₂ ratio under a standard ventilatory setting—a prospective, multicenter validation study. *Intensive Care Med*. 2013;39(4):583–92.
17. Hasper D, et al. Changes in serum creatinine in the first 24 hours after cardiac arrest indicate prognosis: an observational cohort study. *Crit Care*. 2009;13(5):R168.
18. Daviaud F, et al. Blood glucose level and outcome after cardiac arrest: insights from a large registry in the hypothermia era. *Intensive Care Med*. 2014;40(6):855–62.
19. Schuetz P, et al. Serum procalcitonin, C-reactive protein and white blood cell levels following hypothermia after cardiac arrest: a retrospective cohort study. *Eur J Clin Invest*. 2010;40(4):376–81.
20. Andersen LW, et al. In-hospital cardiac arrest: a review. *JAMA*. 2019;321(12):1200–10.
21. Mody P, et al. Targeted temperature management for cardiac arrest. *Prog Cardiovasc Dis*. 2019;62(3):272–8.
22. Sandroni C, et al. ERC-ESICM guidelines on temperature control after cardiac arrest in adults. *Intensive Care Med*. 2022;48(3):261–9.
23. Sinha P, et al. Development and validation of parsimonious algorithms to classify acute respiratory distress syndrome phenotypes: a secondary analysis of randomised controlled trials. *Lancet Respir Med*. 2020;8(3):247–57.

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